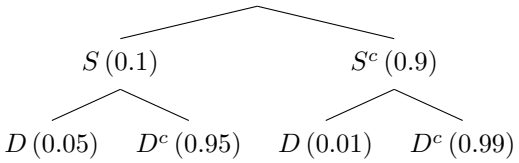
1: Ten percent of a company's life insurance policyholders are smokers. The rest are non-smokers. For each non-smoker, the probability of dying during the year is 0.01. For each smoker, the probability of dying during the year is 0.05. Given that a policyholder died, what is the probability that the policyholder was a smoker?

Solution: Let S be the event that a policyholder is a smoker and D the event that a policyholder dies during the year.



By Bayes' formula we have

$$\begin{aligned} P(S \mid D) &= \frac{P(D \mid S)P(S)}{P(D \mid S)P(S) + P(D \mid S^c)P(S^c)} \\ &= \frac{(0.05)(0.1)}{(0.05)(0.1) + (0.01)(0.9)} \\ &= \frac{0.005}{0.014} \\ &= \boxed{\frac{5}{14} \approx 0.357} \end{aligned}$$

2: An actuary studying the insurance preferences of automobile owners makes the following conclusions:

(i) An automobile owner is twice as likely to purchase collision coverage as disability coverage.

(ii) The event that an automobile owner purchases collision coverage is independent of the event that he or she purchases disability coverage.

(iii) The probability that an automobile owner purchases both collision and disability coverages is 0.15.

Calculate the probability that an automobile owner purchases neither collision nor disability coverage.
A. 0.18 **B. 0.33** C. 0.48 D. 0.67 E. 0.82

Solution: Let C be the event that an owner purchases collision coverage and D the event that an owner purchases disability coverage. From (i) we have $P(C) = 2P(D)$, and from (ii) and (iii) we have

$$\begin{aligned} &P(C \cap D) = P(C)P(D) = 0.15 \\ \Rightarrow &2P(D)^2 = 0.15 \\ \Rightarrow &P(D) = \sqrt{0.075} \approx 0.2739 \\ \Rightarrow &P(C) = 2\sqrt{0.075} \approx 0.5477 \end{aligned}$$

So by inclusion-exclusion

$$\begin{aligned} P(C^c \cap D^c) &= 1 - P(C \cup D) \\ &= 1 - (P(C) + P(D) - P(C \cap D)) \\ &\approx 1 - (0.5477 + 0.2739 - 0.15) \\ &= \boxed{0.3284 \approx 0.33} \end{aligned}$$

and thus $\boxed{\text{B}}$ is the correct answer.

[0] LaTeX code for this document can be found on github here